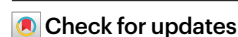


Predicting drug responses of unseen cell types through transfer learning with foundation models

Received: 8 January 2025

Accepted: 3 September 2025

Published online: 03 October 2025



Yixuan Wang^{1,8}, Xinyuan Liu^{1,8}, Yimin Fan¹, Binghui Xie¹, James Cheng¹, Kam Chung Wong², Peter Cheung³, Irwin King¹ & Yu Li^{1,4,5,6,7} ✉

Drug repurposing through single-cell perturbation response prediction provides a cost-effective approach for drug development, but accurately predicting responses in unseen cell types that emerge during disease progression remains challenging. Existing methods struggle to achieve generalizable cell-type-specific predictions. To address these limitations, we introduce the cell-type-specific drug perturbation responses predictor (CRISP), a framework for predicting perturbation responses in previously unseen cell types at single-cell resolution. CRISP leverages foundation models and cell-type-specific learning strategies to enable effective transfer of information from control to perturbed states even with limited empirical data. Through systematic evaluation across increasingly challenging scenarios, from unseen cell types to cross-platform predictions, CRISP shows generalizability and performance improvements. We demonstrate CRISP's drug repurposing potential through zero-shot prediction from solid tumor data to sorafenib's therapeutic effects in chronic myeloid leukemia. The predicted anti-tumor mechanisms, including CXCR4 pathway inhibition, are supported by independent studies as an effective therapeutic strategy in chronic myeloid leukemia, aligning with past studies and clinical trials.

Modeling the impact of drug perturbations on transcriptional responses at the single-cell level is crucial for unraveling molecular mechanisms within cells and accelerating drug development^{1,2}. Despite technological advances, traditional drug development faces critical challenges such as high attrition rates and lengthy development timelines^{3–5}. Drug repurposing, which identifies new ways to use approved drugs, offers a promising alternative with lower risks, shorter development times and reduced costs^{6,7}. Among various repurposing approaches, leveraging transcriptomics data provides unique advantages through access to

widely accessible databases, direct measurement of cellular responses and natural integration with pathway analysis⁸.

Recent progress in single-cell perturbation techniques has provided vital tools for examining drug effects on individual cells with high resolution^{9–13}. However, these techniques are expensive and challenging to apply broadly, particularly for predicting drug responses across multiple cellular populations in heterogeneous diseases^{14–17}, limiting their utility in drug development and precision medicine. This has driven the development of machine-learning methods for predicting

¹Department of Computer Science and Engineering, CUHK, Hong Kong SAR, China. ²Department of Mathematics, HKUST, Hong Kong SAR, China.

³Department of Chemical Pathology, CUHK, Hong Kong SAR, China. ⁴The Chinese University of Hong Kong Shenzhen Research Institute, Shenzhen, China. ⁵Institute for Medical Engineering and Science, Massachusetts Institute of Technology, Cambridge, MA, USA. ⁶Wyss Institute for Biologically Inspired Engineering, Harvard University, Boston, MA, USA. ⁷Broad Institute of MIT and Harvard, Cambridge, MA, USA. ⁸These authors contributed equally: Yixuan Wang, Xinyuan Liu. ✉e-mail: liy@se.cuhk.edu.hk

transcriptional perturbation responses^{18–22}. Despite these advancements, existing methods lack the capacity to generalize across different cell types and treatment conditions in a cell-type-specific manner.

Existing approaches fall into two main categories. Disentanglement-based methods (for example, Biolord²⁰ and chemCPA²¹) use variational autoencoder (VAE) frameworks and molecular embeddings to predict responses for unseen drugs, but remain constrained to training cell types with limited zero-shot capabilities. State transformation methods (for example, scGen¹⁸ and CellOT¹⁹) learn mappings from control to perturbed states for different cell types, but require distinct mappings for each treatment and substantial computational resources when scaling to multiple drugs.

Recent single-cell foundation models (scFMs), such as Geneformer²³, scGPT²⁴, scFoundation²⁵ and SCimilarity²⁶, encode biological information using language model frameworks. Trained on large-scale datasets such as the Human Cell Atlas, these scFMs demonstrate promising zero-shot learning capabilities in characterizing diverse cell states. They provide opportunities for cell type generalization in drug response prediction by generating cell-type-specific embeddings while mitigating batch effects^{23,24}; however, direct integration of scFMs with existing methods fails to yield notable performance gains, prompting an investigation into developing more sophisticated frameworks to effectively leverage the rich cellular information encoded in these models.

Here we propose cell-type-specific drug perturbation responses predictor (CRISP), a versatile framework that leverages scFM-powered transfer learning to predict drug responses in unseen cell types. CRISP incorporates both scFMs and chemical pre-trained models within a unified cell-type-specific learning framework, learning transformation maps from control to drug-perturbed gene expression profiles. This design offers enhanced flexibility by eliminating feature alignment requirements across sequencing platforms while enabling zero-shot predictions for unseen drug-cell type combinations without pre-existing training data. The framework facilitates efficient *in silico* drug screening and generalization across unseen drugs, cell types, tissue and patient data.

We conduct comprehensive experiments across eight distinct scenarios of increasing complexity. CRISP consistently outperforms existing approaches, achieving a 24.5% improvement in correlation coefficients for unseen cell-type predictions. Notably, the framework demonstrates promising applications in drug repurposing, identifying Sorafenib's potential therapeutic effects in unseen cell lines through zero-shot transfer learning, which aligns with independent experimental evidence without information leakage. This framework effectively captures cell type similarities and extends cell atlases into the perturbation space, advancing drug response prediction, drug screening and personalized treatment applications.

Results

Cell-type-specific perturbation prediction framework

Drug perturbation responses depend on both molecular properties and cellular context. CRISP addresses this challenge by learning cell type-aware transformation maps that predict drug responses across different cell types, including unmeasured cell types and drugs (see Fig. 1). CRISP leverages scFMs for control state initialization and chemical embeddings for treatment representation. These models, pre-trained on large-scale datasets, encode gene expression profiles into dense embeddings that capture cellular states^{23–25}. Single-cell measurements are inherently unpaired since cell destruction prevents observing individual cells before and after perturbation¹⁹. To address this challenge, CRISP employs a foundation-model-assisted paired control encoder that generates simulated paired control pre-embeddings for each perturbed cell in a cell-type-specific manner. This approach captures inherent variability across cell types while preserving unperturbed state characteristics. Building on chemCPA²¹, CRISP processes paired control and drug pre-embeddings through separate encoding heads

for latent space transformation. The framework incorporates a specialized training strategy with cell-type-specific classifiers and contrastive learning modules in the perturbation decoder. This architecture captures divergent perturbation responses across cell types, which is essential for accurate prediction in heterogeneous populations. The framework's flexible design eliminates feature alignment requirements across sequencing platforms and enables zero-shot predictions for unseen drug-cell type combinations without pre-existing training data. Integration of mechanism of action (MOA) analysis within the predicted drug-cell treatment space provides biological interpretability, connecting molecular perturbations to functional outcomes and enabling deeper investigation of drug impacts from a mechanistic perspective.

Perturbation response prediction across cell types and drugs

We first evaluate generalization capabilities by predicting drug effects on cell types withheld from perturbed training data (Fig. 2a). This set-up assesses each model's ability to predict drug responses in previously unmeasured cell types, addressing critical challenges in drug screening for rare or poorly characterized cells. We compare CRISP against state-of-the-art methods including CellOT¹⁹, scGen¹⁸, Biolord²⁰, chemCPA²¹ and CPA²² on benchmark datasets NeurIPS²⁷ and SciPlex3². CRISP is evaluated on three different splits with distinct withheld cell types from both datasets. Results demonstrate promising generalization capability; in the NeurIPS dataset, CRISP achieves a 41% improvement in the Pearson correlation coefficient of the logarithm of fold change (Pr_d of $\log(\text{FC})$) (Fig. 2b,c). Our approach shows 24.5% higher performance across all held-out combinations (Supplementary Fig. 2). Notably, chemCPA-FM—an adaptation incorporating the same scFM as CRISP in the encoder component—does not yield substantial improvement compared with chemCPA, highlighting the importance of CRISP's cell-type-specific design.

To examine prediction accuracy in detail, we analyze $\log(\text{FC})$ distributions for two anti-tumor drugs in B cells (Fig. 2d). The analysis focuses on idelalisib (PI3K inhibitor) and palbociclib (CDK4/6 inhibitor), both known for high efficacy in B cell growth inhibition^{28,29}. CellOT demonstrates meaningful correlation with ground truth values but shows performance degradation due to outliers. By comparison, CRISP generates predictions that are closely aligned with the ground truth, particularly for differentially expressed (DE) genes. Genome-wide analysis of model performance across different fold-change magnitudes demonstrates that CRISP genuinely predicts perturbation-specific changes rather than systemic artifacts (Supplementary Figs. 7 and 8).

Prediction accuracy is further evaluated through directional analysis of the top 50 DE genes across held-out cell types (Fig. 2e). CRISP demonstrates higher directional accuracy across most held-out cell types, except for T-regulatory cells. The lower performance on T-regulatory cells may be attributed to training data imbalance, where these cells represent only 4.7% of all immune cells; however, CRISP maintains consistent performance in single-cell resolution metrics, suggesting accurate individual cell-level predictions despite population-level challenges (Supplementary Fig. 3d,e). For single-cell resolution analysis, we examine the expression distribution of the top two DE genes in dactolisib-treated CD4+ T cells, comparing control, ground truth and model predictions (Fig. 2f). The analysis reveals substantial up-regulation of ZFAS1 and SNHG6, two long non-coding RNAs linked to tumorigenesis³⁰. CRISP's predicted expression distributions closely match the ground truth distributions at the individual-gene level, highlighting CRISP's capability to capture drug effects at single-cell resolution (refer to Supplementary Fig. 3f,g for further results for the top ten DE genes).

Accurate simulation of cellular responses to unseen drugs is crucial for new drug design. To evaluate this capability under a more challenging scenario, we assess CRISP and chemCPA on the SciPlex3 dataset where both target cell types and drugs are withheld during training (Fig. 2g). The evaluation used nine held-out anti-tumor compounds, which influence cancer cell proliferation through diverse mechanisms.

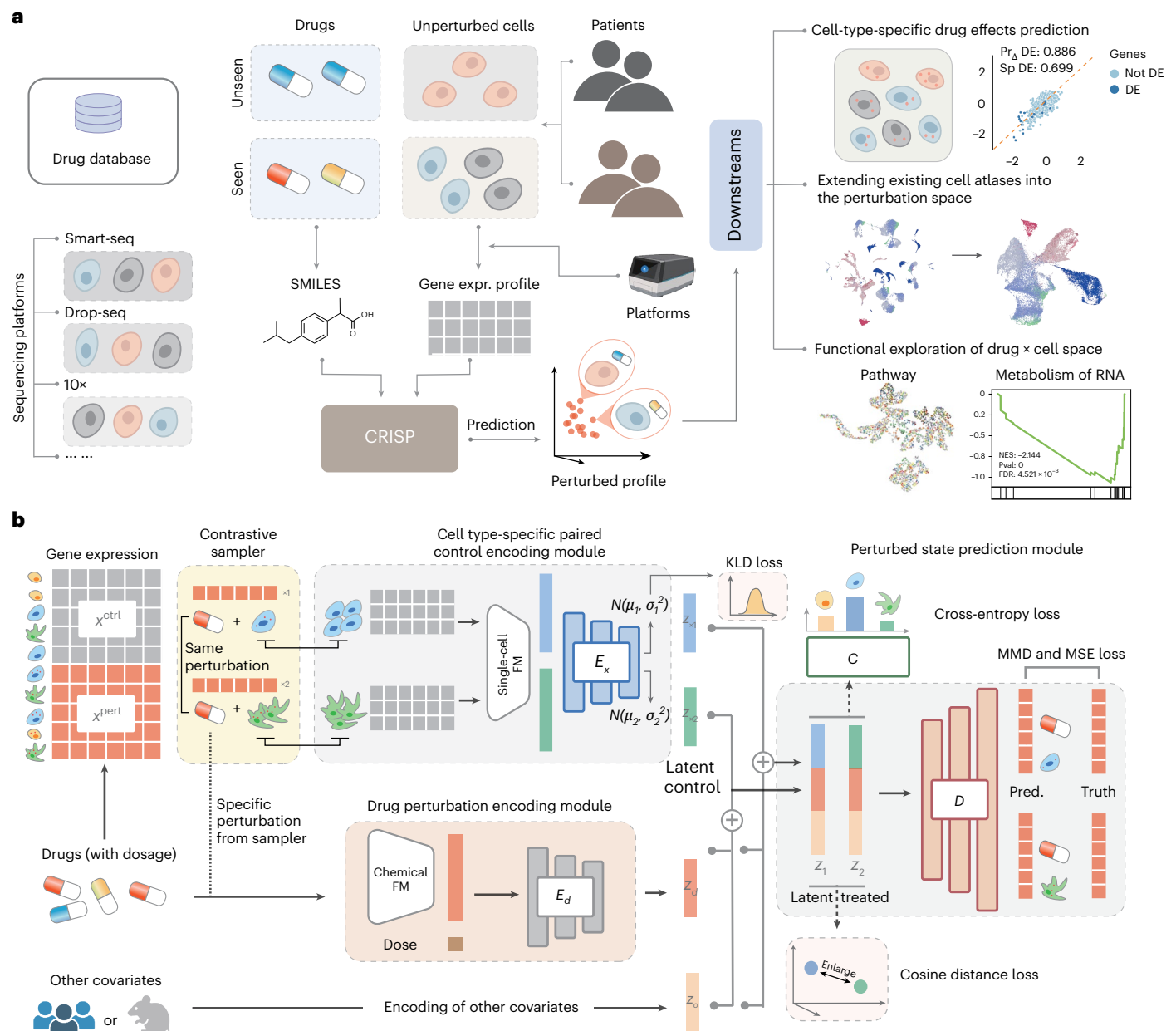


Fig. 1 | Overview of CRISP. a, CRISP is designed to predict drug effects for any treatment combinations, including unexplored new drugs, unmeasured cell types and cells from new patients. Using single-cell foundation models and a well-adapted cell-type-specific framework, CRISP is equipped with generalization capability for various batches and scRNA-seq platforms. The inference process of CRISP only requires treatment conditions and gene expression profiles in the control state, without any feature alignment or annotations. The powerful and flexible framework enables in silico simulation of drug efficacy with superior performance in few- and zero-shot scenarios. The MOA analysis in the predicted drug × cell treatment space is able to investigate the drug's impact from a biological perspective. **b**, A detailed illustration of CRISP's framework. It comprises two modules: the cell-type-specific paired control encoder and

the post-perturbation prediction module. During the training process, to simulate the transfer process from the control to perturb state, CRISP first gets the foundation-model-encoded cell-type-specific control pre-embeddings by averaging across that of paired control samples. Together with the pre-embeddings of drugs, they are passed through two separate encoding heads to get latent embeddings, which are inputs of the perturb state prediction decoder. It is also feasible for CRISP to consider other covariates such as drug dosage and species information. The perturbation decoder contains a cell-type-related contrastive learning module and a cell-type classifier, enabling a better distinction and specificity for different cell types. DE, differentially expressed; FM, foundation model; PCC, Pearson correlation coefficient; Sp, Spearman correlation coefficient. Some icons in **a** and **b** were created using [BioRender.com](#).

Histone deacetylase inhibitors primarily target hematologic malignancies, while aurora kinase inhibitors demonstrate efficacy against solid tumors. The analysis examines predictions for blood cancer (K562) and solid cancer cell lines (A549, MCF7). CRISP quantitatively surpasses the second-best method by an average margin of 17.5% in Pr_{Δ} metrics and 10.3% in Sinkhorn distance, with statistical significance (Fig. 2h).

CRISP also demonstrates superior computational efficiency (Fig. 2i). Although CellOT achieves competitive performance, it requires

separate training for each drug and exhibits 25-fold-longer training times. CRISP's streamlined architecture enables rapid training while maintaining state-of-the-art performance.

Flexible scFM integration and zero-shot prediction capabilities Leveraging existing scFMs, CRISP features a flexible architecture that integrates multiple scFM embeddings with specialized perturbation prediction objectives. We benchmark state-of-the-art scFMs to identify

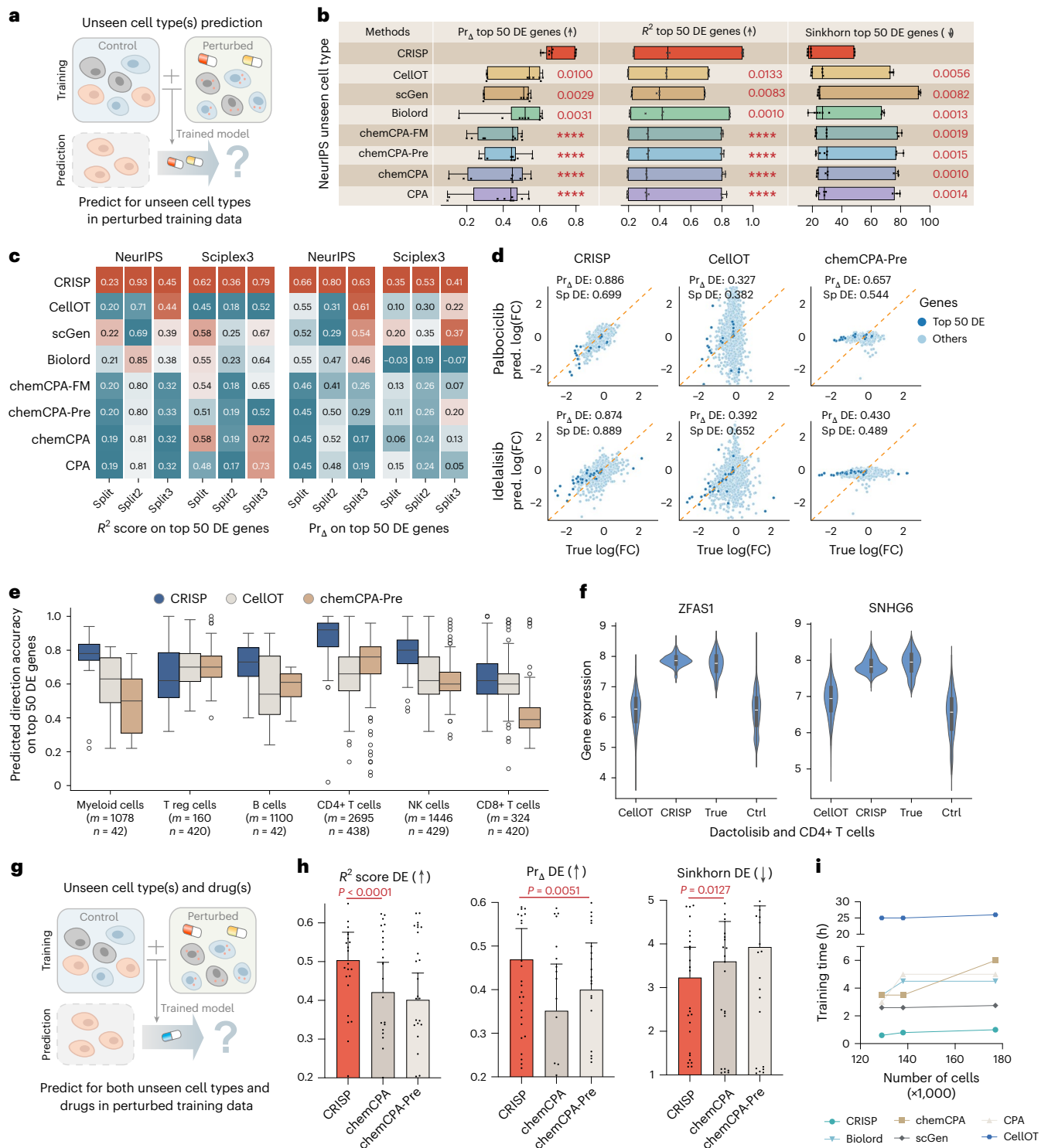


Fig. 2 | Performance of CRISP across unseen cell types. **a**, Illustration of unseen cell-type prediction, with specific cell types withheld during perturbation training. **b**, Performance comparison between CRISP and baseline methods on the NeurIPS dataset. Box plots display the distribution of nine points of three split settings. Boxes represent the 25–75th percentiles, the center line is the median, and the whiskers represent minimum to maximum; all data points are shown. Metric directionality is indicated by arrows. **c**, Split-specific analysis of R² scores and Pr_Δ for different unseen cell types. **d**, Comparison of true versus predicted log(FC) for untrained B-cell treatments. Individual genes are represented as points, with the top 50 DE genes highlighted in dark blue. Pr_Δ and Sp for DE genes are shown in each panel. **e**, Directional prediction accuracy for top 50 DE genes across six held-out cell types. We show the number of cell type–drug combinations (*n*) and the number

of control cells in each cell type group (*m*). Panels **e** and **f** have the same box plot settings as **b**, except that whiskers only extend to data within the 1.5× interquartile range (IQR). **f**, Single-cell expression distribution of two top DE genes in dactolisib-treated CD4+ T cells, comparing control (*n* = 2,598), ground truth (*n* = 760) and model predictions (*n* = 2,695). Cells with expression beyond 2 s.d. were filtered. **g**, Illustration of unseen cell-type(s) and drug(s) prediction scenario. **h**, Performance evaluation of CRISP on the unseen cell-type(s) and drug(s) prediction scenario (36 unique dose–cell type combinations). Error bars represent the 95% confidence interval (CI); all data points are shown. **i**, Computational efficiency comparison across methods using the NeurIPS dataset. We use one-tail paired *t*-test for each statistical analysis. The corresponding significance is annotated with red text (values ≤ 0.0001 are indicated by four stars).

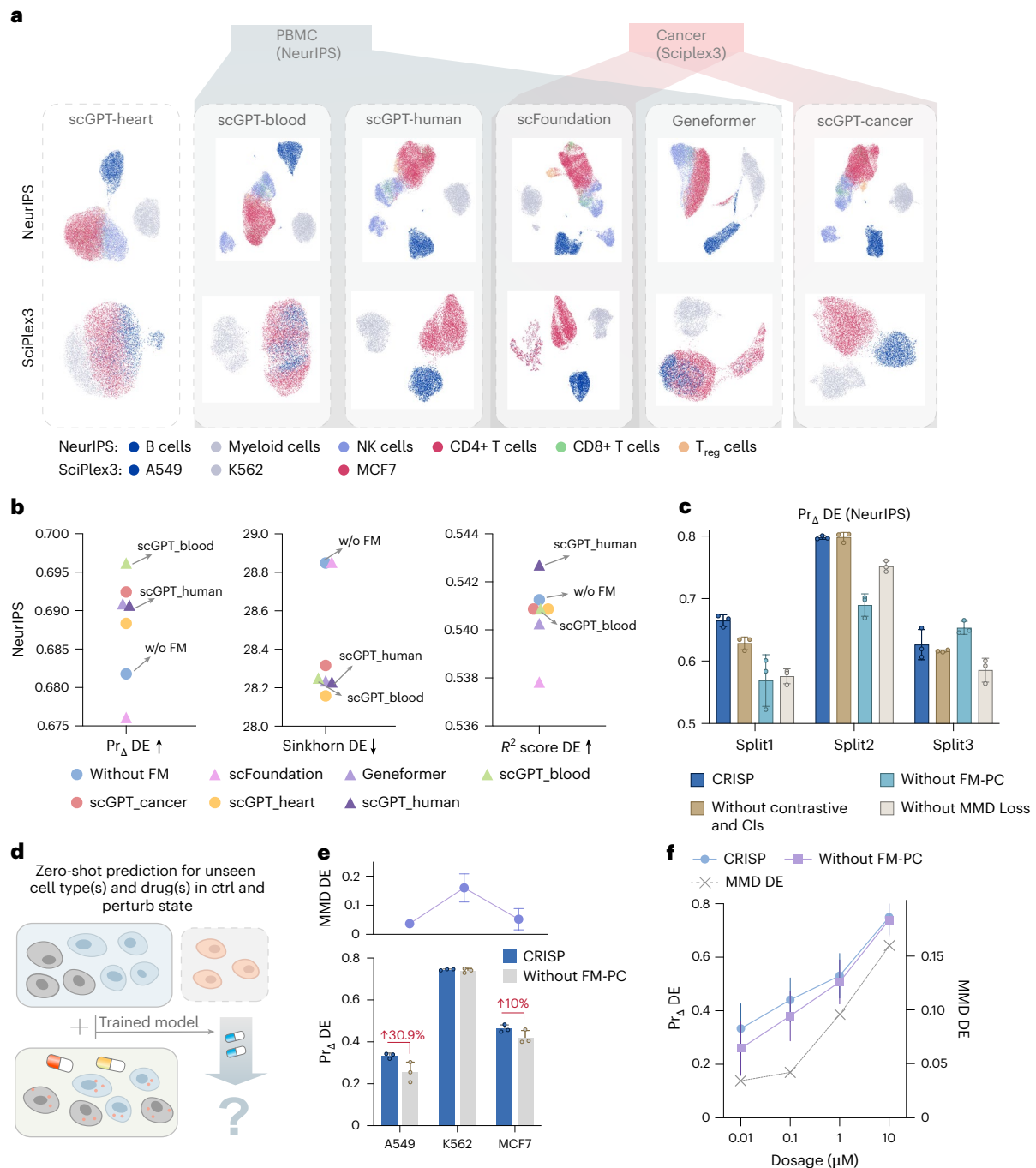


Fig. 3 | Adaptation of scFMs to CRISP for zero-shot drug perturbation response prediction on unseen cell types. a, UMAP visualizations comparing pre-trained cell embeddings from different scFMs applied to two datasets: NeurIPS (PBMCs) and SciPlex3 (three cancer cell lines). The colored boxes indicate the scFM pretraining data coverage for the target perturbation cell types. **b**, Performance comparison of different scFMs on the NeurIPS dataset, with triangle symbols marking scFMs that cover required cell types. Baseline results without scFM integration are included. **c**, Ablation analysis using the Pr_Δ metric to evaluate the impact of removing various key components: FM-assisted paired control encoding module (without FM-PC), contrastive learning and cell-type classification losses (without contrastive and CIs) and MMD loss (without

MMD loss). Error bar shows the s.d. of three replications per unseen cell types around the mean (center). **d**, Conceptual illustration of zero-shot prediction for cell types and drugs excluded from both control and perturbed training data. **e**, Comparative analysis showing MMD of DE genes between control and perturbed states per split (upper plot), and CRISP's performance in zero-shot prediction across both new cell types and drugs (bottom). Each point represents an experiment replication, with three points for each column. Error bar shows the s.d. around mean (center). **f**, Line graph showing the correlation between drug dosage and prediction performance. There are 81 drug-cell type combinations (points) per column. Error bar represents the 95% CI around mean (center).

optimal models for prediction across unseen cell types. As various scFM training datasets contain heterogeneous cell-type compositions—with some models excluding specific treated cell types—careful consideration of cell type overlap between pretraining and target datasets is crucial for model selection. Pre-embeddings of untreated

cells from various scFMs are visualized in Fig. 3a, with color-coded boxes indicating correspondence between perturbed datasets and relevant scFMs. The overlap-FM-derived embeddings demonstrate clear cell-type clustering across datasets, where overlap-FMs denote scFMs with matching cell types in pretraining data. For instance, scGPT-cancer

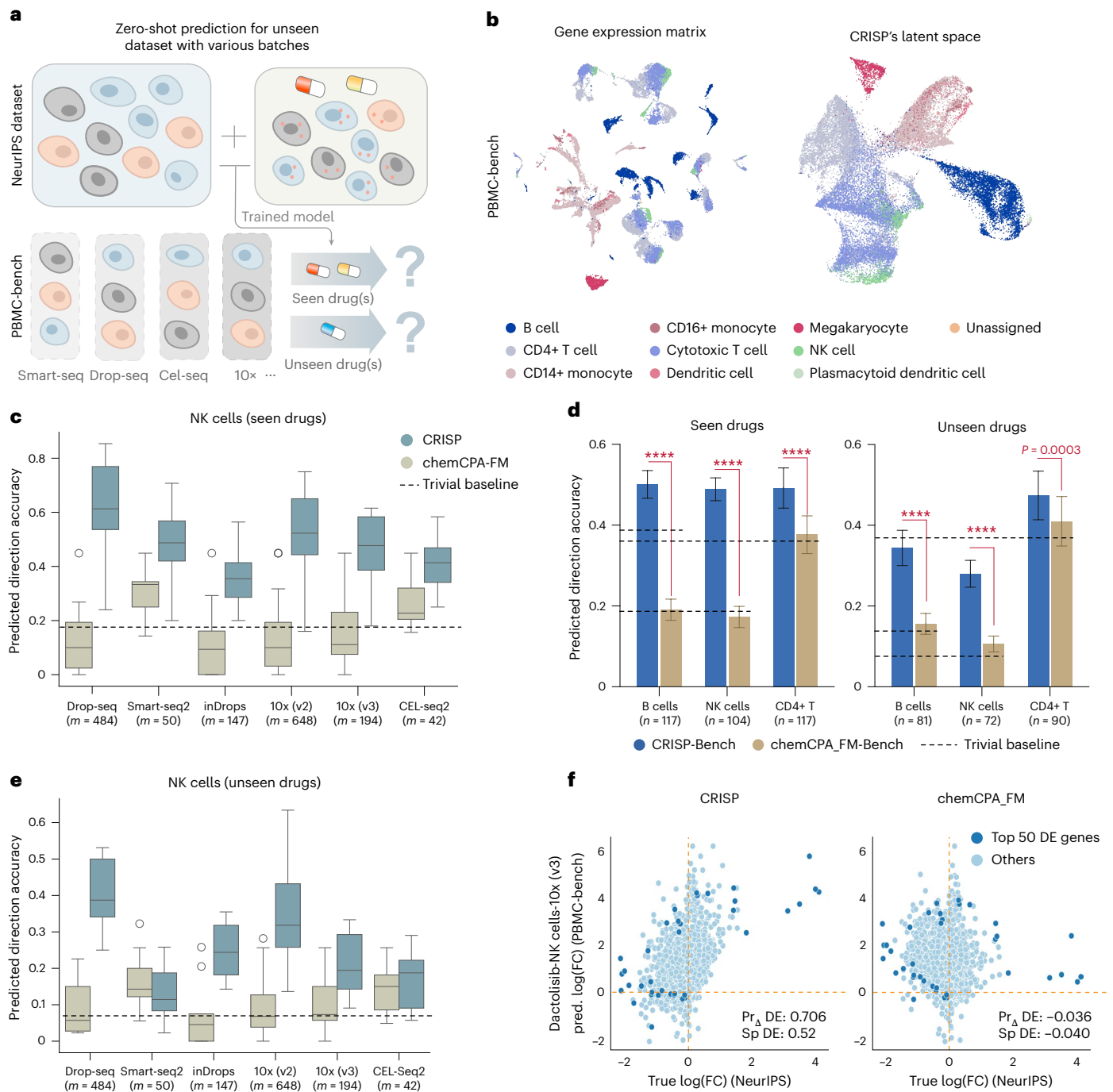


Fig. 4 | Zero-shot prediction with CRISP across batches of other datasets. **a**, Illustration of zero-shot prediction on the PBMC-bench dataset across different sequencing platforms using the trained CRISP model. **b**, UMAP visualizations of the original gene expression matrix and CRISP's latent embeddings. **c,e**, Predicted direction accuracy on NK cells across various batches on seen (**c**) and unseen drugs (**e**), respectively. Boxes represent the 25–75th percentiles, the center line is the median, whiskers represent the minimum to maximum value within 1.5× of the IQR, and outliers beyond this range are shown. Each point corresponds to one of the 13 unseen drugs or 9 seen drugs.

The number of control cells in each group (m) is noted. **d**, Direction accuracy of predicted genes' fold change across multiple cell types from different batches in the PBMC-bench dataset, shown for both seen and unseen treatment conditions. Each point shows one drug–batch combination; the data size (n) are noted under each column. The statistical significance of one-tail paired t -test is annotated in red text (values ≤ 0.0001 are indicated by four stars). Error bar shows the 95% CI around the mean. **f**, Scatter plot of predicted versus true log(FC) of measured genes for the dactolisib-NK cells-10x (v3) combination. The corresponding Pr_{Δ} and Sp of the top 50 DE genes are annotated.

provides enhanced characterization on the SciPlex3 dataset compared to scGPT-heart and scGPT-blood. Performance comparison of CRISP models with different scFMs is presented in Fig. 3b and Supplementary Fig. 4, with overlap-FMs marked with triangles. Overlap-scGPT models demonstrate notable performance, comparing favorably to both Geneformer and scFoundation. Our approach maintains consistent

performance across varying scFM choices, suggesting stability in FM integration. Ablation studies (Fig. 3c and Supplementary Fig. 4) reveal contributions from individual CRISP modules. The FM-assisted paired control encoding module improves performance substantially, increasing Pr_{Δ} by 0.07 and 0.12 for NeurIPS and SciPlex3 datasets, respectively. Cell-type-specific regularization contributes approximately 75%

improvement in Pr_{Δ} on the SciPlex3 dataset. Furthermore, maximum mean discrepancy (MMD) loss integration enhances CRISP's performance by 11% for the NeurIPS dataset.

The practical application of perturbation prediction models often involves entirely new sample batches. We design a rigorous experiment excluding specific cell types and drugs from both unperturbed and perturbed training data to evaluate zero-shot prediction capabilities (Fig. 3d). With its cell type encoding-free architecture and scFM integration, CRISP enables flexible application to new cell types with promising zero-shot prediction. As other methods are not applicable to this challenging scenario, we evaluate CRISP's zero-shot predictions against its ablated version (without FM-PC) on the SciPlex3 dataset, assessing Pr_{Δ} of DE genes across held-out cell lines (Fig. 3e). Within each split, the held-out drugs are the same as the prediction setting in Fig. 2g,h, with four treatment dose levels. Performance variations across cancer cell lines probably reflect the distinct characteristics of held-out tumor samples relative to training data and cell-type-specific drug responses. Drug dosage substantially affects perturbation outcomes². Given CRISP's enhanced performance in cell groups with larger treatment MMD, we examine dosage-dependent performance across four dosage levels in the SciPlex3 dataset (Fig. 3f). The analysis reveals increasing Pr_{Δ} values with higher dosages, indicating that CRISP's predictive capability scales with perturbation strength. This trend corresponds with increasing MMD of differentially expressed genes at higher doses. Notably, CRISP achieves effective control-to-perturbed state transformation in the harder cases with lower drug doses, or less responsive cell lines, showing enhanced sensitivity to subtle perturbation effects through effective scFMs integration.

Extending existing cell atlases into the perturbation space

Beyond within-dataset predictions, we investigate CRISP's zero-shot performance across datasets with large batch effects or different tissues. Batch effects between different scRNA-seq datasets pose a major challenge to the prediction quality of computational methods. Current drug perturbation prediction tools often rely on transfer learning from bulk perturbation data to specific datasets, lacking robust zero-shot prediction capabilities²¹. CRISP addresses this limitation through large pre-trained scFMs and cell-type-specific training procedures. We evaluate this capability using PBMC-bench, an unperturbed dataset containing peripheral blood mononuclear cell (PBMC) samples from diverse sequencing platforms, as well as the PC9 dataset, which contains non-small cell lung cancer (NSCLC) samples. The CRISP model trained on the NeurIPS PBMC dataset is applied to predict responses for both seen and unseen drugs on these platform-specific samples (Fig. 4a and Extended Data Fig. 1), creating a rigorous zero-shot evaluation scenario.

We analyze both original gene expression profiles and CRISP's latent embeddings of PBMC-bench through UMAP visualizations (Fig. 4b). The discrepancies within original profiles are attributed to substantial batch effects between different platforms. CRISP's learned latent space demonstrates enhanced cell-type separation and organization, indicating effective encoding of biological information while minimizing technical artifacts. We use chemCPA-FM as a baseline for cross-batch drug perturbation predictions, adapting it to use the same scFM as CRISP in the encoder component. A trivial baseline using average NeurIPS–PBMC differences is included. Without ground-truth post-perturbation data in PBMC-bench, we evaluate predictions by calculating directional accuracy against the same cell types in the NeurIPS dataset, assuming consistent perturbation effects within identical cell types across different batches. The evaluation focuses on B cells, natural killer (NK) cells, and CD4+ T cells that are present in both datasets. Analysis of NK cells reveals CRISP's improved performance over chemCPA-FM in predicting gene expression directional changes, despite sharing identical scFM encoders (Fig. 4c). This advantage extends to unseen drugs (Fig. 4e), where ten drugs are held out from

the training dataset, including tyrosine kinase inhibitors (crizotinib, foretinib, R428), PI3K inhibitor (idelalisib), and others with distinct mechanisms of action. CRISP demonstrates higher accuracy in predicting directions of $\log(FC)$ across different platforms except Smart-Seq2, validating CRISP's effectiveness in zero-shot prediction scenarios.

The predictive performance of CRISP for unseen drugs improves with larger datasets but decreases under conditions with fewer cells, as evidenced in Smart-Seq2 and CEL-Seq2 experiments (Fig. 4e). This could stem from increased within-population variance in zero-shot predictions when dealing with unseen drugs in smaller sample sizes. From the cell type perspective, CRISP demonstrates broad predictive capabilities across multiple cell types and experimental batches in the PBMC-bench dataset, consistently outperforming baseline models (Fig. 4d). For detailed assessment, we examine prediction for dactolisib treatment of NK cells sequenced using the 10x platform (Fig. 4f). The scatter plot reveals strong correlation between predicted and true $\log(FC)$, demonstrating that CRISP effectively captures gene-level perturbation patterns. In this cross-dataset context, correlation more meaningfully assesses prediction quality than magnitudes of $\log(FC)$, which naturally differ between datasets due to batch effects. We further visualize the overall atlas of CRISP-inferred perturbation space containing both PBMC and cancer cells (Extended Data Fig. 1), demonstrating well-characterized organization even in cross-tissue and cross-dataset zero-shot scenarios, providing a potential solution for transferring normal cell atlases into perturbation space for drug screening.

Functional analysis for potential drug repurposing

To evaluate CRISP's alignment with established biological pathways and drug actions, we examine two zero-shot prediction scenarios where CRISP has no past exposure to either the specific cell type or patient and drug. Leveraging the large-scale unperturbed single-cell expression data in PBMC-bench dataset, we visualize the comprehensive landscape of CRISP's drug $\times$ cell space (Fig. 5a–c), encompassing both in-distribution and zero-shot predictions. The cell type-based visualization (Fig. 5c) reveals clear stratification patterns, indicating that CRISP effectively captures drug effects across different cell types in a context-dependent manner.

Although tyrosine kinase inhibitors such as imatinib mesylate have revolutionized chronic myelogenous leukemia (CML) treatment, tyrosine kinase inhibitor resistance remains a substantial challenge³¹, necessitating alternative therapeutic options. Sorafenib, a clinically approved multi-kinase inhibitor targeting tumor cell proliferation and angiogenesis, is primarily used to treat solid tumors³². Given its mechanism of action, we hypothesize that sorafenib might be effective against hematologic malignancies such as CML. To test this hypothesis, we apply CRISP to predict sorafenib's effects on K562 CML cells in a zero-shot manner. Gene set enrichment analysis (GSEA) of these predicted outcomes reveals that CRISP accurately predicts sorafenib's potential anti-tumor effects in CML through multiple biologically relevant mechanisms (Fig. 5a,d and Supplementary Fig. 6b). CRISP captures the down-regulation of mitochondrial genes MT-RNR1 and MT-RNR2, as well as RNA metabolism, which aligns with past reports that sorafenib can induce apoptosis in hepatocellular carcinoma cells by disrupting mitochondrial function^{33–35}. The model also predicts down-regulation of the P53 pathway, leukocyte *trans*-endothelial migration, and the CXCR4 pathway in CML cells. The CXCR4 finding is particularly relevant as this signaling pathway facilitates interactions between CML cells and the bone marrow microenvironment, representing a critical factor in cell survival and drug resistance³⁶. The clinical relevance of our predictions is supported by sorafenib's completed phase II clinical trial in imatinib-resistant CML patients³⁷, demonstrating sorafenib's potential efficacy against CML despite some side effects. We conduct a comprehensive drug screening experiment on all 188 drugs in the SciPlex3 dataset against K562 cells, showing high precision in identifying effective anti-cancer drugs (Extended Data Fig. 2).

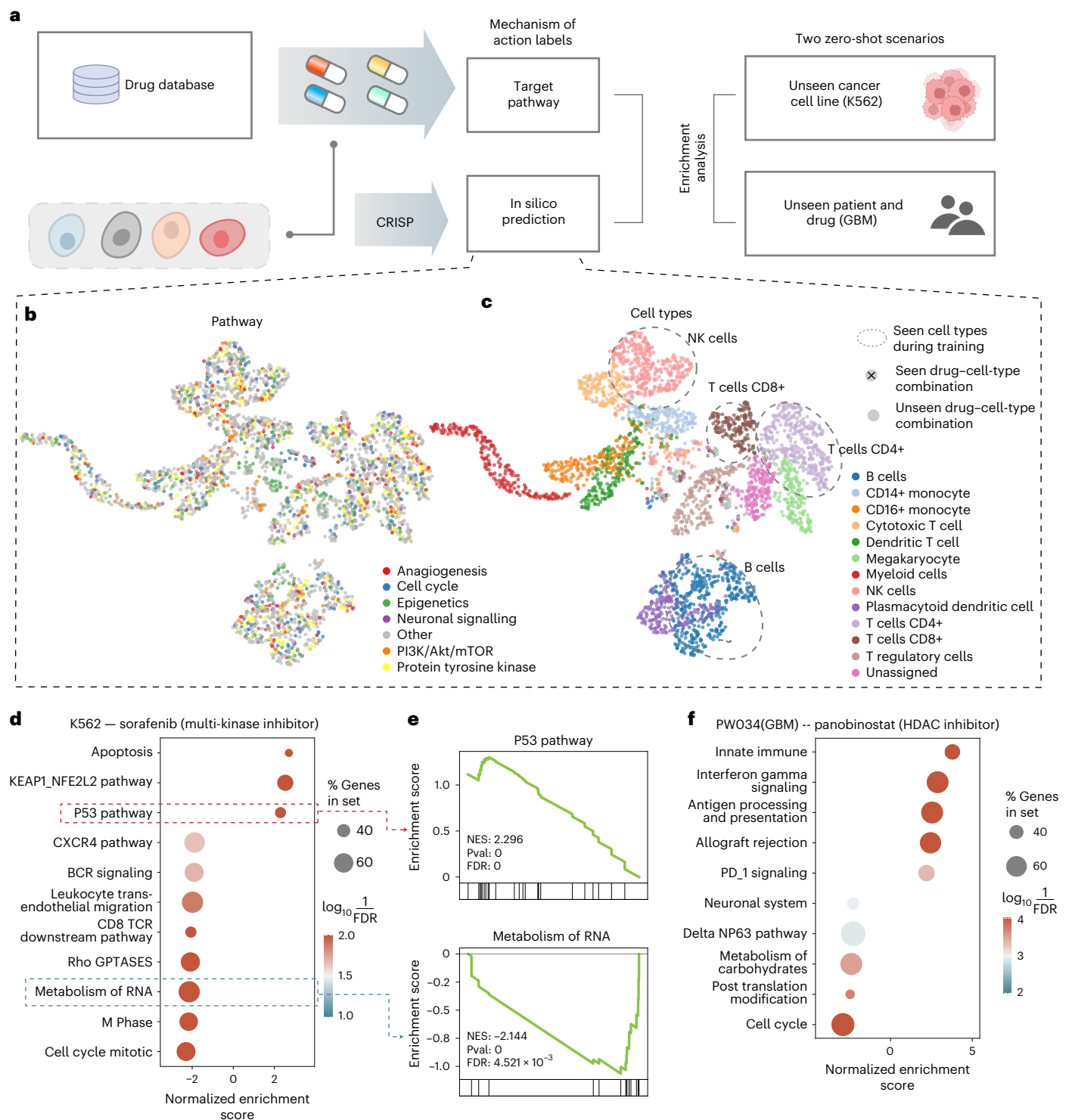


Fig. 5 | MOA analysis of CRISP's in silico prediction. **a**, The enrichment analysis of CRISP's in silico prediction on two zero-shot scenarios: unseen cell types (K562) and unseen patients and drugs. The MOA information and target of each drug are acquired from various drug-related databases. **b, c**, The landscape of cell-type-specific drug effect prediction, where each point represents a combination of cell type and drug. They are colored with the related pathway (**b**) and cell type (**c**) respectively. The UMAP is generated from the perturbed expression profiles predicted by CRISP, enabling visualization of how different drug-cell type combinations cluster in the gene expression space. **d**, Visualization of enrichment analysis of the prediction on unseen cell line K562

with sorafenib treatment. The size of the circle shows the ratio of overlapped regulated genes, and the depth of color shows the statistical significance level. GSEA was performed between control and predicted perturbed groups using a two-sided signal-to-noise statistic with 1,000 phenotype-based permutations; and FDR q-values are used to assess significance. The same GSEA statistical settings were used for panels **e** and **f**. **e**, Two function terms are emphasized from GSEA results: positive enriched P53 pathway and negative enriched metabolism of RNA. Their detailed GSEA plots are shown. **f**, The GSEA results of zero-shot prediction of panobinostat-treated GBM cells from patient PW034, where drug and patient information are both unseen during training.

To evaluate cross-patient generalization, we apply CRISP to predict unseen drug panobinostat's effect on glioblastoma (GBM) cells from unseen patients and perform GSEA analysis (Fig. 5f and Supplementary Fig. 6). The histone deacetylase inhibitor panobinostat has shown potential in GBM treatment by promoting tumor cell death³⁸. Despite no exposure to panobinostat and GBM cells of PW034, CRISP's zero-shot prediction reveals positive enrichment on immune response and negative enrichment on cell cycle and neural system. The predicted down-regulation of MDM2 gene is known to be associated with cell proliferation and neural system development^{39,40}. CRISP precisely simulates patient responses to unseen drugs in a flexible manner, substantially enhancing the convenience and speed of personalized treatment simulation and drug screening.

These results emphasize CRISP's ability to flexibly screen drug and cell type combinations in a cell-type-specific way, enabling acceleration of drug discovery timelines and reduction of experimental expenses. The captured drug-specific mechanisms of action validate its predictions against known biological effects, building confidence in CRISP's potential as a tool for both drug repurposing and personalized treatment.

Discussion

We introduce CRISP, a flexible and robust scFM-assisted framework designed for cell-type-aware drug perturbation prediction at single-cell resolution. As a cell-type-specific framework for predicting drug responses for unseen cell types, CRISP demonstrates strong capability to distinguish between cell types, regardless of their similarity or divergence; however, our evaluation revealed some limitations that warrant discussion. Although CRISP maintains consistent performance in single-cell resolution metrics for T-regulatory cells, its population-level evaluation metrics are affected by the limited sample sizes available for these rare cell types. Furthermore, although CRISP demonstrates cross-platform zero-shot prediction capabilities that exceed trivial baselines, the accuracy of approximately 0.5 indicates that there is considerable room for improvement. Currently, our cross-platform evaluation is limited to direction accuracy metrics due to the lack of ground truth cross-platform perturbation data, but explicitly modeling technical factors during training could greatly enhance performance across different experimental platforms. Despite performing initial evaluations on cross-tissue prediction, our method's robustness across divergent tissues and experimental conditions requires further enhancement.

Several future directions remain to be explored. First, although the chemical model encoder allows CRISP to predict unknown drugs, a more powerful chemical encoding module needs development. Integrating additional drug information beyond simplified molecular input line entry system (SMILES) strings and RDKit-encoded chemical properties may enhance drug encoding and predictive performance. The influence of drug–drug interactions also requires further research, as common disease treatments often involve multi-drug combinations where one drug may impact the efficacy or resistance of another⁴¹, yet no methods that specifically model the transcriptional effects of combination drug perturbations at single-cell resolution currently exist^{20,22}, partly due to the current lack of comprehensive single-cell datasets featuring drug combinations. Second, recent strategies such as neural optimal transport and causal inference have shown promise^{19,42}. However, the optimal transport learning strategy overlooks cell type divergence during single-cell state mapping and requires separate training for different treatments, making it inflexible and time-consuming. Therefore, beyond the current cell-type-aware training strategies used by CRISP, conditional optimal transport maps could be applied to CRISP's framework to further improve single-cell perturbation prediction ability. Third, despite the integration of scFMs, a more comprehensive foundational model for large-scale drug screening is still needed. It could be pre-trained from multi-modality sources, including

chemical perturbation, drug molecular and unperturbed single-cell sequencing data. Finally, although CRISP identifies the potential of Sorafenib in chronic myelogenous leukemia treatment, comprehensive pathway and drug repurposing analysis on more existing drugs from larger drug databases is still required for medical application and drug discovery. Moreover, the side effects of proposed drugs still require further investigation through clinical trials, and the outcomes of CRISP should be considered as preliminary references only.

The flexible and powerful CRISP framework enables accurate simulation of biological shifts in specific drugs and targeted cell types. With its enhanced cell-type-specific predictions, CRISP further enables the understanding of treatment effects on lesions with specific cellular compositions, which is valuable in practical applications. Furthermore, CRISP has potential for accelerating drug repurposing by identifying potential applications for investigational drugs across various cell types or cancer lines. Overall, CRISP's framework paves the way for developing high-resolution, personalized therapeutic recommendation systems, and substantially advances drug treatment discovery in medicine.

Methods

Implementation details of CRISP

Drug perturbation outcomes are influenced by multiple factors, including treatment conditions, cellular context in the control state, cell type and individual metadata such as species. The core architecture of CRISP consists of a VAE, E , integrated with a pre-trained scFM. It is designed to generate accurate and robust simulations of drug responses across diverse drugs and cell types. In the inference phase, a key component is the scFM-assisted cell-type-specific paired control encoder. CRISP learns the posterior distribution of latent embedding $\mathbf{z}_{x_i}$, conditioned on the control-state cellular gene expression $\mathbf{x}_i^{\text{ctrl}}$, denoted as $q(\mathbf{z}_{x_i}|\mathbf{x}_i^{\text{ctrl}})$. The control cell embedding $\mathbf{z}_{x_i}$ is sampled and concatenated with drug treatment embedding $\mathbf{z}_d$ and additional covariate embedding $\mathbf{z}_o$ to generate a latent treated embedding $\mathbf{z}_i^{\text{treat}}$. Cell type specificity in $\mathbf{z}_i^{\text{treat}}$ is enforced through cell type classification and contrastive learning. In summary, CRISP comprises two principal components: a scFM-assisted paired control encoder and a cell-type-specific perturbation prediction decoder module. The detailed training algorithm can be found in Supplementary Algorithm 1.

scFM-assisted paired control encoder. Unlike vanilla encoder architecture with multi-layer perceptron (MLP), CRISP uses scFM to initialize sparse expression profiles to dense embeddings. In the default setting we leverage the pre-trained scGPT model $\mathbf{G} : \mathbb{R}^M \rightarrow \mathbb{R}^h$ to map control gene expression vector $\mathbf{x}_i^{\text{ctrl}}$ to pre-trained space, denoted as $\mathbf{g}_{x_i} \in \mathbb{R}^h$. Due to the lack of paired observed control and perturbed data, CRISP generates a synthetic paired control pre-embedding for each perturbed sample. To achieve this, a paired control group k_i is first identified, consisting of unperturbed samples that share the same covariate attributes as the perturbed sample. The paired control embedding $\mathbf{g}_{x_i}$ is then constructed by averaging the embeddings $\mathbf{g}_{x_j}$ of all members within group k_i . If we refer to the perturbation condition as p_i , the process could be represented by:

$$\mathbf{g}_{x_i} = \begin{cases} \mathbf{G}(\mathbf{x}_i), & p_i = \text{control}, \\ \frac{1}{|k_i|} \sum_{j \in k_i} \mathbf{G}(\mathbf{x}_j), & p_i = \text{perturbed}. \end{cases} \quad (1)$$

Unlike CRISP, other autoencoder-based approaches such as chemCPA use the perturbed state as input and employ a disentanglement learning strategy. Our choice to use the control profile as input is based on two key considerations. First, the paired control state captures heterogeneous genetic features of the cellular context while excluding perturbation-related information, thereby preserving the biological relevance of the cellular context and minimizing noise during training.

Second, the foundational model is typically trained on single-cell data from the control state, establishing it as a standard and widely recognized reference point.

Considering the potential loss of heterogeneity within cell type groups due to the averaging process of embeddings, we apply the reparameterization trick of VAE to generate diverse samples from the posterior distribution, simulating the sampling process. Passing paired control embedding $\mathbf{g}_{x_i}$ through gene encoder $\mathbf{E}_x: \mathbb{R}^h \rightarrow \mathbb{R}^l$, the posterior distribution of latent control embedding $q(\mathbf{z}_{x_i} | \mathbf{x}_{x_i}^{\text{ctrl}})$ is inferred, where $\mathbf{E}_x$ outputs parameters of $\mathcal{N}(\mu, \sigma)$, and samples the latent control embedding $\mathbf{z}_{x_i}$ from it.

As for the treatment condition, we keep the treatment embedding part the same as that for chemCPA, which uses molecular model RDKit⁴³ $\mathbf{G}_d: \mathcal{G} \rightarrow \mathbb{R}^m$ to encode molecular representation $\mathbf{d} \in \mathcal{G}$ to drug embedding $\mathbf{g}_d \in \mathbb{R}^m$. CRISP then encodes the drug molecular and its dosage s to the perturbation latent embedding $\mathbf{z}_{d_i} \in \mathbb{R}^l$, by molecular encoder $\mathbf{E}_d$ and dosage-specific encoder $\mathbf{E}_s$:

$$\mathbf{z}_{d_i} = \mathbf{E}_d(\mathbf{g}_{d_i}) \times \mathbf{E}_s([\mathbf{g}_{d_i}, s]^T), \text{ with } \mathbf{g}_{d_i} = \mathbf{G}_d(\mathbf{d}_i). \quad (2)$$

Together with the sampled unperturbed cellular latent embedding $\mathbf{z}_{x_i}$ we mentioned above, there are $2 + t$ terms in the latent space where $t \geq 0$ is the number of additional covariates except cell type. The treated latent embedding could be represented as:

$$\mathbf{z}_i^{\text{treat}} = [\mathbf{z}_{x_i}, \mathbf{z}_{d_i}, \mathbf{z}_{o_i}]^T, \quad (3)$$

where $\mathbf{z}_{o_i}$ is the latent embedding of other covariates. We normally keep $\mathbf{z}_{o_i}$ empty in our experiments.

Cell-type-specific prediction module. Decoder $\mathbf{D}: \mathbb{R}^{(2+t)l} \rightarrow \mathbb{R}^M$ is an MLP and takes the treated latent embedding $\mathbf{z}_i^{\text{treat}}$ to predict the perturbed gene expression profile $\hat{\mathbf{x}}_i \in \mathbb{R}^M$. At the same time, we add an extra constrain to $\mathbf{z}_i^{\text{treat}}$ with cell type labels to enlarge differentiation between divergent cell types.

Rather than using one-hot encoding for the cell-type covariate, we employ cell-type classification and contrastive learning as regularization techniques to enhance the learning of a cell-type-specific latent space. Furthermore, our scFM-assisted cellular context encoding inherently incorporates cell-type information. This framework enables in silico predictions in the absence of explicit cell type labels, while simultaneously preserving the distinctiveness of each cell type group. The cell type classification is implemented with an extra MLP as a classifier appended to the treated latent embedding $\mathbf{z}_i^{\text{treat}}$. To perform contrastive learning, we add a contrastive sampler at the beginning of the model. For each target perturbed cell, one negative cell with a different cell type but the same perturbation condition is randomly sampled. We perform contrastive learning by enlarging the cosine distance from the negative one ($\mathbf{z}_j^{\text{treat}}$). In summary, the regularization terms in the loss function of the cell type classifier and contrastive learning can be formulated as:

$$\mathcal{L}_{\text{reg}} = \alpha (\mathbf{CE}(\mathbf{C}(\mathbf{z}_i^{\text{treat}}), c) + 0.1 \times \text{Cosine loss}(\mathbf{z}_i^{\text{treat}}, \mathbf{z}_j^{\text{treat}})), \quad (4)$$

where $\mathbf{C}$ is the cell type classifier, c is the cell type label and α is the coefficient of cell-type-specific regularization.

Training objective of CRISP. The objective function (evidence lower bound) of CRISP's VAE framework can be divided into three parts: the prediction loss measured by negative log likelihood, the Kullback–Leibler divergence (KLD) and the regularization term. First, the prediction loss is measured by L2 (MSE) and MMD to minimize the error between predicted and observed post-perturbation gene expression profiles. Given the inherently high-dimensional nature of scRNA-seq data, it is observed that only a limited subset of genes exhibit noticeable changes

under perturbations. To depict the disparity between perturbed and control state expressions, we refined the vanilla L2 loss to prioritize the top T differential expression genes, ranked based on the absolute value of log fold change (log FC). Due to the unpaired observed data, $\text{MMD}(\hat{\mathbf{x}}_i^{\text{treat}}, \mathbf{x}_i^{\text{treat}})$ is calculated within each minibatch and used as an addition to MSE. It measures the divergence between two distributions, instead of computing for each pair of samples individually. We also add two constraint items as regularization of treated latent embeddings: cross-entropy loss of cell type classification and penalty of contrastive learning using cosine distance. In practice, we use CosineEmbeddingLoss() function from the PyTorch package to implement contrastive learning loss.

Application of foundation models. We evaluate three scFMs in our experiments: Geneformer²³, scGPT²⁴ and scFoundation²⁵. To streamline the process, we first extract and store the inferred cell embeddings from each scFM independently, avoiding repeated inference during each CRISP run. For computational efficiency, we keep the parameters of the pre-trained foundational model fixed, updating only the appended VAE component during training to optimize time and memory resources. In the default setting, we use scGPT-blood as a pre-encoder for NeurIPS, and scGPT-cancer as a pre-encoder for SciPlex3 and GBM. We apply scGPT-human to all samples for cross-tissue prediction.

Drug treatment encoding. Building on the new drug perturbation prediction method, chemCPA, CRISP pre-encodes drug treatment by using a chemical foundation model RDKit⁴³. RDKit is a widely used cheminformatics library in Python that offers various tools for encoding and analyzing small molecules. It uses standard representations of small molecules, the SMILES strings, as input. In practice, the descriptastorus package that wraps RDKit and other descriptor generators is utilized to generate molecular encoding using multiple molecular properties. Given the SMILES string of a specific drug, it is encoded by RDKit into 201-dimension space. The feature dimension containing NA is removed, and the normalized embedding matrix is used for initialization of the drug encoder.

Calculation of KLD. Kullback–Leibler divergence is a measure of information theory that quantifies how one probability distribution Q diverges from a second reference probability distribution P . It is often used in machine learning, particularly for training variational autoencoders and other models where distributions are compared. As we assume the prior distribution of Q is $\mathcal{N}(0, 1)$, the formula for KLD is:

$$D_{\text{KL}}(P|Q) = D_{\text{KL}}(\mathcal{N}(\mu, \sigma^2) | \mathcal{N}(0, 1)) = -\frac{1}{2} \sum_{i=1}^d (1 + \log(\sigma^2) - \mu^2 - \sigma^2) \quad (5)$$

where d is the dimensionality of hidden space, μ and σ^2 is the mean and variance of posterior distribution $P = \mathcal{N}(z; \mu, \sigma^2)$

Calculation of MMD. The MMD loss is a statistical measure used to quantify the difference between the means of two distributions in a reproducing kernel Hilbert space. The MMD loss is computed using a radial basis function (RBF) kernel, which allows for the calculation of the MMD between two sets of samples X and Y in a reproducing kernel Hilbert space. Given two sets of samples, the MMD loss is defined as:

$$\text{MMD}(X, Y) = \mathbb{E}_{\mathbf{x}, \mathbf{x}}[k(\mathbf{x}, \mathbf{x})] - 2\mathbb{E}_{\mathbf{x}, \mathbf{y}}[k(\mathbf{x}, \mathbf{y})] + \mathbb{E}_{\mathbf{y}, \mathbf{y}}[k(\mathbf{y}, \mathbf{y})] \quad (6)$$

where $k(\mathbf{x}, \mathbf{y})$ is the RBF kernel between samples $\mathbf{x}$ and $\mathbf{y}$, $\mathbb{E}_{\mathbf{x}, \mathbf{x}}$ denotes the expectation over pairs of samples from X . As the distribution of our samples is Gaussian shape and CRISP incorporating multiple bandwidth multipliers, the RBF kernel can be expressed as:

$$k(\mathbf{x}, \mathbf{y}) = \sum_{m=1}^M \exp\left(-\frac{\|\mathbf{x} - \mathbf{y}\|^2}{\sigma^2 \times \text{multiplier}_m}\right) \quad (7)$$

where x and y are the data points, $\|x - y\|^2$ is the squared Euclidean distance between the points, σ^2 is the bandwidth parameter that controls the spread of the kernel, and is calculated by

$$\sigma^2 = \frac{1}{n^2 - n} \sum_{i=1}^n \sum_{j \neq i}^n \|X_i - Y_j\|^2 \quad (8)$$

where n is the number of samples, M is the number of kernels (set to 5 in CRISP) and multiplier _{m} corresponds to the m th multiplier defined in the RBF class. The bandwidth multipliers are factors that scale the base bandwidth to create a range of bandwidths for different kernels. They are defined as:

$$\text{multipliers} = \text{mul_factor}^{\left(\frac{k-M}{2}\right)} \quad \text{for } k = 0, 1, \dots, M-1 \quad (9)$$

where (mul_factor) is a predefined scaling factor (set to 2 in CRISP). The multiplier is computed for each kernel by raising the multiplication factor to the power of k shifted by half the number of kernels (M). This creates a series of bandwidths centered around the base bandwidth, allowing the model to capture various scales of similarity between data points.

Datasets

Four publicly available scRNA-seq perturbation datasets, and one unperturbed scRNA-seq dataset under CC-BY 4.0, are used to perform our experiments: the NeurIPS dataset from NeurIPS competition of single-cell perturbation prediction²⁷, the SciPlex3 dataset containing single-cell measurements of three cancer cell lines and 187 drugs², the GBM dataset containing different patients, the PC9 dataset containing drug-treated non-small cell lung cancer samples, and unperturbed dataset PBMC-bench with various sequencing platforms. Information on the potential targets and MOA for the drugs investigated in this study was obtained from multiple established drug databases, including DrugBank⁴⁴, PubChem⁴⁵, LINCS data portal⁴⁶ and the annotation file from the SciPlex3 dataset².

NeurIPS dataset. The NeurIPS dataset²⁷ measures the perturbation response across 18,211 genes of 240,059 cells from PBMC to 147 compositions, including 144 drugs with dosage as 1 μ M, dimethylsulfoxide (DMSO) with dosage as 14.1 μ M as negative control and two drugs (dabrafenib and belinostat) with dosage as 0.1 μ M as a positive control. It contains six cell types, including five lymphoid cell types: CD4+ T cells, CD8+ T cells, NK cells, T-regulatory cells, B cells, and one myeloid cell type. For B cells and myeloid cells, only part of the drug perturbation data is accessible. In the experiments, we treat all drugs except DMSO as the treatment group and samples with DMSO as the control group. Moreover, to be more comprehensive, the perturbation prediction is performed across all 18,211 gene features without the gene subset.

SciPlex3 dataset. SciPlex3 data² contain the perturbation response of 581,777 cells from three cancer cell lines: A549 human lung adenocarcinoma, MCF7 breast adenocarcinoma and K562 chronic myelogenous leukemia cells. There are 187 drug treatments, and single-cell transcriptome measurement is performed in 58,347 genes in total.

GBM dataset. The GBM dataset⁴⁷ contains measurements for 155,377 cells from glioblastoma tissue with six drug treatments across 60,725 genes. Samples of this dataset come from nine different patients, we use GBM to perform the prediction across different unseen patients.

PC9 dataset. The PC9 dataset⁴⁸ contains single-cell transcriptome measurements for NSCLC. Here we use two sub-datasets (GSE134839 and GSE160244) for evaluation, which contains crizotinib- and erlotinib-treated NSCLC samples. There are 4,000 highly variable genes left after filtering.

PBMC-bench Dataset. The PBMC-bench dataset⁴⁹ contains scRNA-seq data of 31,021 PBMC cells in an unperturbed state from seven sequencing methods: Drop-seq, inDrops, Seq-Well, 10x Chromium (v2), 10x Chromium (v2) A, 10x Chromium (v2) B, 10x Chromium (v3), CEL-Seq2 and Smart-seq2. Samples come from nine cell types. The scRNA-seq is measured across 33,694 genes.

Preprocessing. The gene expression matrix of the above-mentioned three datasets all undergo the same preprocessing to be normalized as a Gaussian distribution. To start with, all of the samples are normalized by total read counts across all genetic features; log transformations are then applied later. Regarding the high dimensionality of single-cell RNA sequencing of SciPlex3 and GBM datasets, we perform feature selection to only preserve the top 5,000 highly variable genes to do training and evaluation.

Differential expression genes selection. The DE genes are those that show statistically significant differences in expression level between the control state and the perturbed state. We identify top 50 DE genes for each perturbation group defined by cell type and treatment condition using the single-cell data analysis tool Scanpy⁵⁰ and sorted according to absolute fold change value. We use t -test and Benjamini-Hochberg corrected method for DE genes identification. These DE gene sets will be used in the calculation of MSE_DE loss and in the evaluation of prediction performance.

MOA annotation. The potential targets and MOA of the investigated drugs are cataloged in multiple drug databases, including DrugBank⁴⁴, PubChem⁴⁵, LINCS data portal⁴⁶ and the annotation file from the SciPlex3 dataset².

Experimental details

Evaluation metrics. To comprehensively assess performance across various scenarios, we use the R^2 value and the PCC of log(FC) (Pr_Δ) for average evaluations, along with Sinkhorn distance to assess prediction quality at the single-cell resolution. Given that our primary focus is on perturbation-induced changes in DE genes, the evaluation metrics are calculated based on the top 50 DE genes. In practice, we use the PearsonR function from the scipy package⁵¹, the R2Score function from the TorchMetrics package⁵² and SampleLoss function (for sinkhorn distance calculation) from the GeomLoss package⁵³.

R^2 score. The R^2 score—or coefficient of determination—is a statistical metric used to evaluate the accuracy of a regression model. It measures the proportion of the variance in the dependent variable that is predictable from the independent variables. The formula for the R^2 score is as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (10)$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the observed values and n is the number of samples. The R^2 score is used as one of the evaluation metrics of the mean of predicted single-cell gene expression profiles in a specific treatment combination group. It ranges between 0 and 1, with values closer to 1 indicating a better model fit. There are some cases in which R^2 can be negative, and we set R^2 to 0 when this happens.

Pearson correlation coefficient. The PCC measures the linear relationship between two variables. Its value ranges from -1 to 1 , where 1 indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship and 0 indicates no linear relationship. The formula for calculating the PCC between two variables X and Y is:

$$\text{PCC} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (11)$$

where X_i and Y_i are the individual data points for variables X and Y , $\bar{X}$ and $\bar{Y}$ are the means of X and Y , respectively, and n is the number of data points.

The PCC is widely used to assess the strength and direction of the linear association between two continuous variables. We applied PCC to measure the relationship between the predicted log(FC) and observed log(FC), where the variables X and Y represent the mean of predicted and observed log(FC) for each gene, respectively.

Sinkhorn distance. Sinkhorn distance is a modified version of the Wasserstein distance, often used in optimal transport problems. It calculates the minimal cost required to transform one probability distribution into another, while incorporating an entropic regularization term. This regularization improves computational efficiency, making it suitable for high-dimensional data and large-scale machine-learning applications. The Sinkhorn distance between two discrete probability distributions, $\mu = \{\mu_i\}_{i=1}^n$ and $\nu = \{\nu_j\}_{j=1}^m$, with a cost matrix $C \in \mathbb{R}^{n \times m}$ (typically the pairwise Euclidean distance matrix between points), is defined as:

$$\text{Sinkhorn}_\epsilon(\mu, \nu) = \min_{\pi \in U(\mu, \nu)} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} C_{ij} + \epsilon \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} \log(\pi_{ij}) \quad (12)$$

where π is the transport plan matrix that maps μ to ν , $U(\mu, \nu)$ is the set of valid transport plans that satisfy the marginal constraints $\pi \mathbf{1} = \mu$ and $\pi^T \mathbf{1} = \nu$, ϵ is a positive regularization parameter that controls the trade-off between the transportation cost and the entropy. The entropy term $\epsilon \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} \log(\pi_{ij})$ encourages smoother and more computationally efficient solutions compared to the classic Wasserstein distance. During evaluation, Sinkhorn distance is used to measure the distance between predicted and true gene expression profiles in single-cell resolution, where μ represents the group of predicted profiles and ν represents the group of ground truth. The transport plan matrix π is iteratively optimized during the Sinkhorn distance calculation until it satisfies the marginal constraints of the source and target distributions. Once the final transport plan matrix π is obtained, the Sinkhorn distance can be computed directly⁵⁴.

Direction accuracy. The direction accuracy indicates the accuracy of predicted direction (up- or down-regulated) within the top 50 DE genes. It is calculated from the average of predicted and observed log(FC) per treatment group. In particular, the input of the accuracy calculation is the bool variables that 1 means up-regulation, and 0 means down regulation. The direction accuracy is formulated below,

$$\text{Acc}(\hat{y} - \hat{x}, y - x) = \frac{\sum_{i=1}^N \mathbb{1}(\text{sign}(\hat{y}_i - \hat{x}_i) = \text{sign}(y_i - x_i))}{N} \quad (13)$$

where y_i and $\hat{y}_i$ are the averaged actual and predicted perturbed states of gene i , respectively; x_i is the averaged control state of gene i ; $\hat{x}$ and x are the same if the prediction is performed on the same dataset as the reference; and $\mathbb{1}(\cdot)$ is an indicator function that returns 1 if the signs match and 0 otherwise. The direction accuracy is calculated for each treatment-cell type combination.

Trivial baseline. To establish a trivial baseline for log fold change (log(FC)) direction accuracy, we compute the actual log(FC) as the difference between the average perturbed state $\mathbf{Y}$ and the average control state $\mathbf{X}$ for each cell type-drug condition (c, d) . The baseline prediction, instead of using drug-specific perturbations, averages all perturbed states within a cell type. The direction accuracy is then

evaluated by comparing the direction consistency between the predicted and actual log(FC):

$$\text{Accuracy}_{\text{baseline}}^c = \frac{1}{|D|} \sum_{d \in D} \text{Acc}((\bar{\mathbf{Y}}(c) - \bar{\mathbf{X}}(c)), (\bar{\mathbf{Y}}(c, d) - \bar{\mathbf{X}}(c))), \quad (14)$$

where $\bar{\mathbf{Y}}(c, d)$ and $\bar{\mathbf{X}}(c)$ represent the average expression values in the perturbed and control states in the corresponding conditions, respectively. Acc represents the direction accuracy of the given prediction and observed log(FC). We calculate the trivial baseline for each cell type separately.

Gene set enrichment analysis. We perform GSEA in this study using GSEAPy Python package⁵⁵. The curated gene sets hallmark and C2-Canonical Pathway are used for GSEA.

Experiment settings. From the perspective of types of held-out information, our experiments can be broadly divided into two scenarios: (1) predict new cell types, and (2) predict new drugs and cell types. As for the source of held-out OOD samples, it can come from the same dataset with different cell types or treatments (Figs. 2 and 3), from a different dataset with a large batch effect (Fig. 4), from different tissues or even different patients. We also try two different strategies for predicting incoming data: (1) incorporating unperturbed data into training, and (2) directly predicting using trained CRISP (zero-shot case). We investigated a total of eight different experimental settings, with detailed illustrations and explanations provided in Supplementary Fig. 1.

Hyperparameters setting. CRISP hyperparameters, such as the learning rate, batch size, and most of the loss item coefficients, are consistent across cases. We use the Adam optimizer with a selected learning rate of 1×10^{-3} from $[1 \times 10^{-2}, 1 \times 10^{-3}, 1 \times 10^{-4}]$. The batch size is set to 128 from [128, 256, 512]. We assign the coefficient of DE-focus MSE loss to 0.75 from [0, 0.5, 0.75, 1]; and the coefficient for cell-type-specific regularization α to 1 from [0, 0.1, 1, 10] (we set this to 0.1 when predicting for unseen drugs). The weight decay and drop out ratio are defined as 1×10^{-7} and 0.2, respectively. As the SciPlex3 dataset contains about two times more samples than the NeurIPS dataset, making it prone to overfitting in OOD splits, we train it for fewer epochs (25) compared with NeurIPS (50). It is recommended to adjust the number of training epochs based on the size of the training dataset when early stopping is not used; the user can also use the default value (50). Furthermore, the coefficient of MMD loss is adjusted according to the value of MSE loss during training. As a result, coefficient of MMD is 1×10^{-4} for SciPlex3 and GBM, 0.1 for NeurIPS. The consistency of hyperparameters and generalization ability of CRISP enables it to be applied conveniently.

The average performance across three distinct cell type-OOD splits of the NeurIPS dataset is presented in Supplementary Table 1 to assess the sensitivity of different hyperparameters. We find that CRISP maintains robust performance across a reasonable range of hyperparameter settings, with the MMD loss coefficient being one of the most sensitive parameters that needed to be properly balanced against the L2 reconstruction loss (MSE). It is also worth noting that the learning rate should not be too low, and a value of 0.001 is a good choice.

Statistics and reproducibility

The one-sided, paired t -test is conducted for a statistical test of comparison results. P -value less than 0.05 is considered to be significant. The sample size is defined by the available samples in each dataset. For the statistical analysis in Fig. 2f, cells with gene expression values beyond ± 2 s.d. from the mean were excluded to mitigate the influence of outliers. All other analyses include the complete set of data points without any exclusion. We apply different seeds to initialize the model's weights before training. The investigators are not blinded to allocation during experiments and outcome assessment. The evaluation process is

consistent across all baselines and our model. All of the experiments are conducted on a single GPU node of the internal cluster equipped with NVIDIA GeForce RTX 4090 and 24,564 MiB of memory and Python 3.9.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All of the datasets used in this research can be accessed publicly. The single-cell drug perturbation dataset SciPlex3² and the GBM dataset⁴⁷ can be obtained via GEO using access codes [GSM4150378](#) and [GSE148842](#). The NeurIPS dataset²⁷ was obtained from [NeurIPS competition](#); the PC9 dataset⁴⁸ can be obtained via GEO with access code [GSE149383](#). The unperturbed PBMC-Bench data⁴⁹ can be found [here](#). Detailed description of these datasets can be found in the ‘Datasets’ section of the Methods. Source data are provided with this paper.

Code availability

The open-source codes of CRISP can be found via <https://github.com/ml4bio/CRISP> and on Zenodo via <https://doi.org/10.5281/zenodo.16713259> (ref. 56).

References

- Van de Sande, B. et al. Applications of single-cell RNA sequencing in drug discovery and development. *Nat. Rev. Drug Discov.* **22**, 496–520 (2023).
- Srivatsan, S. R. et al. Massively multiplex chemical transcriptomics at single-cell resolution. *Science* **367**, 45–51 (2020).
- Zang, C. et al. High-throughput target trial emulation for Alzheimer’s disease drug repurposing with real-world data. *Nat. Commun.* **14**, 8180 (2023).
- Rodriguez, S. et al. Machine learning identifies candidates for drug repurposing in Alzheimer’s disease. *Nat. Commun.* **12**, 1033 (2021).
- Charpignon, Marie-Laure et al. Causal inference in medical records and complementary systems pharmacology for metformin drug repurposing towards dementia. *Nat. Commun.* **13**, 7652 (2022).
- Elhasasna, H. et al. A drug repurposing screen identifies fludarabine phosphate as a potential therapeutic agent for N-MYC overexpressing neuroendocrine prostate cancers. *Cells* **11**, 2246 (2022).
- Yu, M. et al. Deep learning large-scale drug discovery and repurposing. *Nat. Comput. Sci.* **4**, 600–614 (2024).
- Pushpakom, S. et al. Drug repurposing: progress, challenges and recommendations. *Nat. Rev. Drug Discov.* **18**, 41–58 (2019).
- Stubington, Michael J. T., Rozenblatt-Rosen, O., Regev, A. & Teichmann, S. A. Single-cell transcriptomics to explore the immune system in health and disease. *Science* **358**, 58–63 (2017).
- Papalexi, E. & Satija, R. Single-cell rna sequencing to explore immune cell heterogeneity. *Nat. Rev. Immunol.* **18**, 35–45 (2018).
- Norman, T. M. et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science* **365**, 786–793 (2019).
- Replogle, J. M. et al. Mapping information-rich genotype-phenotype landscapes with genome-scale perturb-SEQ. *Cell* **185**, 2559–2575 (2022).
- Doench, J. G. Am i ready for CRISPR? A user’s guide to genetic screens. *Nat. Rev. Genet.* **19**, 67–80 (2018).
- Nakamura, M., Gao, Y., Dominguez, A. A. & Qi, L. S. CRISPR technologies for precise epigenome editing. *Nat. Cell Biol.* **23**, 11–22 (2021).
- Hanna, R. E. & Doench, J. G. Design and analysis of crispr–cas experiments. *Nat. Biotechnol.* **38**, 813–823 (2020).
- He, B. et al. Combination therapeutics in complex diseases. *J. Cell. Mol. Medicine* **20**, 2231–2240 (2016).
- Jagadeesh, K. A. et al. Identifying disease-critical cell types and cellular processes by integrating single-cell RNA-sequencing and human genetics. *Nat. Genet.* **54**, 1479–1492 (2022).
- Lotfollahi, M., Wolf, F. A. & Theis, F. J. scGen predicts single-cell perturbation responses. *Nat. Methods* **16**, 715–721 (2019).
- Bunne, C. et al. Learning single-cell perturbation responses using neural optimal transport. *Nat. Methods* **20**, 1759–1768 (2023).
- Piran, Z., Cohen, N., Hoshen, Y. & Nitzan, M. Disentanglement of single-cell data with biolord. *Nat. Biotechnol.* **42**, 1678–1683 (2024).
- Hetzel, L. et al. Predicting cellular responses to novel drug perturbations at a single-cell resolution. In *Proc. 36th International Conference on Neural Information Processing System* Vol. 35, 26711–26722 (2022).
- Lotfollahi, M. et al. Predicting cellular responses to complex perturbations in high-throughput screens. *Mol. Syst. Biol.* e11517 (2023).
- Theodoris, C. V. et al. Transfer learning enables predictions in network biology. *Nature* **618**, 616–624 (2023).
- Cui, H. et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nat. Methods* **21**, 1470–1480 (2024).
- Hao, M. et al. Large-scale foundation model on single-cell transcriptomics. *Nat. Methods* **21**, 1481–1491 (2024).
- Heimberg, G. et al. A cell atlas foundation model for scalable search of similar human cells. *Nature* **638**, 1085–1094 (2025).
- NeurIPS 2023: Single-cell Perturbation Prediction* (NeurIPS, 2023); <https://www.kaggle.com/competitions/open-problems-single-cell-perturbations/data>
- Burger, J. A. & Okkenhaug, K. Idelalisib—targeting PI3Kδ in patients with B-cell malignancies. *Nat. Rev. Clin. Oncol.* **11**, 184–186 (2014).
- Petroni, G., Formenti, S. C., Chen-Kiang, S. & Galluzzi, L. Immunomodulation by anticancer cell cycle inhibitors. *Nat. Rev. Immunol.* **20**, 669–679 (2020).
- Wu, X. et al. Recent advances in dual PI3K/mTOR inhibitors for tumour treatment. *Front. Pharmacol.* **13**, 875372 (2022).
- Shah, N. P. et al. Sequential ABL kinase inhibitor therapy selects for compound drug-resistant BCR-ABL mutations with altered oncogenic potency. *J. Clin. Investigat.* **117**, 2562–2569 (2007).
- Wilhelm, S. et al. Discovery and development of sorafenib: a multikinase inhibitor for treating cancer. *Nat. Rev. Drug Discov.* **5**, 835–844 (2006).
- Kanehisa, M. & Goto, S. Kegg: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* **28**, 27–30 (2000).
- Paech, F. et al. Mechanisms of mitochondrial toxicity of the kinase inhibitors ponatinib, regorafenib and sorafenib in human hepatic HepG2 cells. *Toxicology* **395**, 34–44 (2018).
- Ek, F. et al. Sorafenib and nitazoxanide disrupt mitochondrial function and inhibit regrowth capacity in three-dimensional models of hepatocellular and colorectal carcinoma. *Sci. Rep.* **12**, 8943 (2022).
- Weisberg, E. et al. Inhibition of CXCR4 in CML cells disrupts their interaction with the bone marrow microenvironment and sensitizes them to nilotinib. *Leukemia* **26**, 985–990 (2012).
- Sorafenib Completed Phase 2 Trials for Chronic Myelogenous Leukemia (CML) Treatment* (DrugBank, accessed 3 August 2025); https://go.drugbank.com/drugs/DB00398/clinical_trials?condition=DBCND0040386&phase=2&purpose=treatment&status=completed

38. Lee, E. Q. et al. Phase II study of panobinostat in combination with bevacizumab for recurrent glioblastoma and anaplastic glioma. *Neuro-oncology* **17**, 862–867 (2015).
39. Xiong, S., Van Pelt, C. S., Elizondo-Fraire, A. C., Liu, G. & Lozano, G. Synergistic roles of Mdm2 and Mdm4 for p53 inhibition in central nervous system development. *Proc. Natl Acad. Sci. USA* **103**, 3226–3231 (2006).
40. Konopleva, M. et al. MDM2 inhibition: an important step forward in cancer therapy. *Leukemia* **34**, 2858–2874 (2020).
41. Scripture, C. D. & Figg, W. D. Drug interactions in cancer therapy. *Nat. Rev. Cancer* **6**, 546–558 (2006).
42. Dong, M. et al. Causal identification of single-cell experimental perturbation effects with CINEMA-OT. *Nat. Methods* **20**, 1769–1779 (2023).
43. Landrum, G. et al. *RDKit: Open-Source Cheminformatics Software* (RDKit, 2016).
44. Knox, C. et al. Drugbank 6.0: the drugbank knowledgebase for 2024. *Nucleic Acids Res.* **52**, D1265–D1275 (2024).
45. Kim, S. et al. Pubchem 2023 update. *Nucleic Acids Res.* **51**, D1373–D1380 (2023).
46. Stathias, V. et al. Lincs data portal 2.0: next generation access point for perturbation-response signatures. *Nucleic Acids Res.* **48**, D431–D439 (2020).
47. Zhao, W. et al. Deconvolution of cell type-specific drug responses in human tumor tissue with single-cell RNA-seq. *Genome Med.* **13**, 82 (2021).
48. Aissa, A. F. et al. Single-cell transcriptional changes associated with drug tolerance and response to combination therapies in cancer. *Nat. Commun.* **12**, 1628 (2021).
49. Ding, J. et al. Systematic comparison of single-cell and single-nucleus RNA-sequencing methods. *Nat. Biotechnol.* **38**, 737–746 (2020).
50. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* **19**, 15 (2018).
51. Virtanen, P. et al. Scipy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**, 261–272 (2020).
52. Detlefsen, NickiSkafte et al. TorchMetrics-measuring reproducibility in PyTorch. *J. Open Source Softw.* **7**, 4101 (2022).
53. Feydy, J. et al. Interpolating between optimal transport and MMD using sinkhorn divergences. In *Proc. 22nd International Conference on Artificial Intelligence and Statistics* 2681–2690 (PMLR, 2019).
54. Cuturi, M. Sinkhorn distances: lightspeed computation of optimal transport. In *Proc. Advances in Neural Information Processing Systems* Vol. 26 (NIPS, 2013).
55. Fang, Z., Liu, X. & Peltz, G. GSEAPy: a comprehensive package for performing gene set enrichment analysis in Python. *Bioinformatics* **39**, btac757 (2023).
56. Wang, Y. et al. Predicting drug responses of unseen cell types through transfer learning with foundation models. *Zenodo* <https://doi.org/10.5281/zenodo.16713259> (2025).
57. Crino, L. et al. Initial phase II results with crizotinib in advanced ALK-positive non-small cell lung cancer (NSCLC): profile 1005. *J. Clin. Oncol.* **29**, 7514 (2011).

Acknowledgements

This study was supported by the Chinese University of Hong Kong (CUHK; award numbers 4937025, 4937026, 5501517, 5501329 and SHIAE BME-p1-24 to Y.L.) and the IdeaBooster Fund (IDBF23ENG05

and IDBF24ENG06 to Y.L.) and partially supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region (Hong Kong SAR), China (project number CUHK 24204023 and 14208525 to Y.L.) and grants from the Innovation and Technology Commission of the Hong Kong SAR, China (project number GHP/O65/21SZ, ITS/247/23FP and PRP/O33/24FX to Y.L.). The work described in this paper was partially supported by the Research Grants Council of the Hong Kong Special Administrative Region, China (CUHK 2410072, RGC R1015-23 to I.K.). This research was also funded by the Research Matching Grant Scheme at CUHK (award numbers 8601603 and 8601663 to Y.L.) from the Research Grants Council, Hong Kong SAR, China. We acknowledge Research Grants Council of Hong Kong Collaborative Research Fund: C6036-21 to P.P.C. and Young Collaborative Research Grants: C4002-24 to Y.L. and P.P.C.

Author contributions

Y.L. conceived the research. Y.W. led the project and designed the experiments. X.L. ran baseline models, prepared detailed tutorials, and maintained the codebase. Both Y.W. and X.L. contributed to model architecture and algorithm implementation, as well as interpretation of the results. Y.W. and X.L. organized and processed the publicly available data. Y.W. and X.L. visualized the experimental results. Y.L. and I.K. supervised the research. Y.W. and X.L. wrote the paper. All authors read and approved the final paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at

<https://doi.org/10.1038/s43588-025-00887-6>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s43588-025-00887-6>.

Correspondence and requests for materials should be addressed to Yu Li.

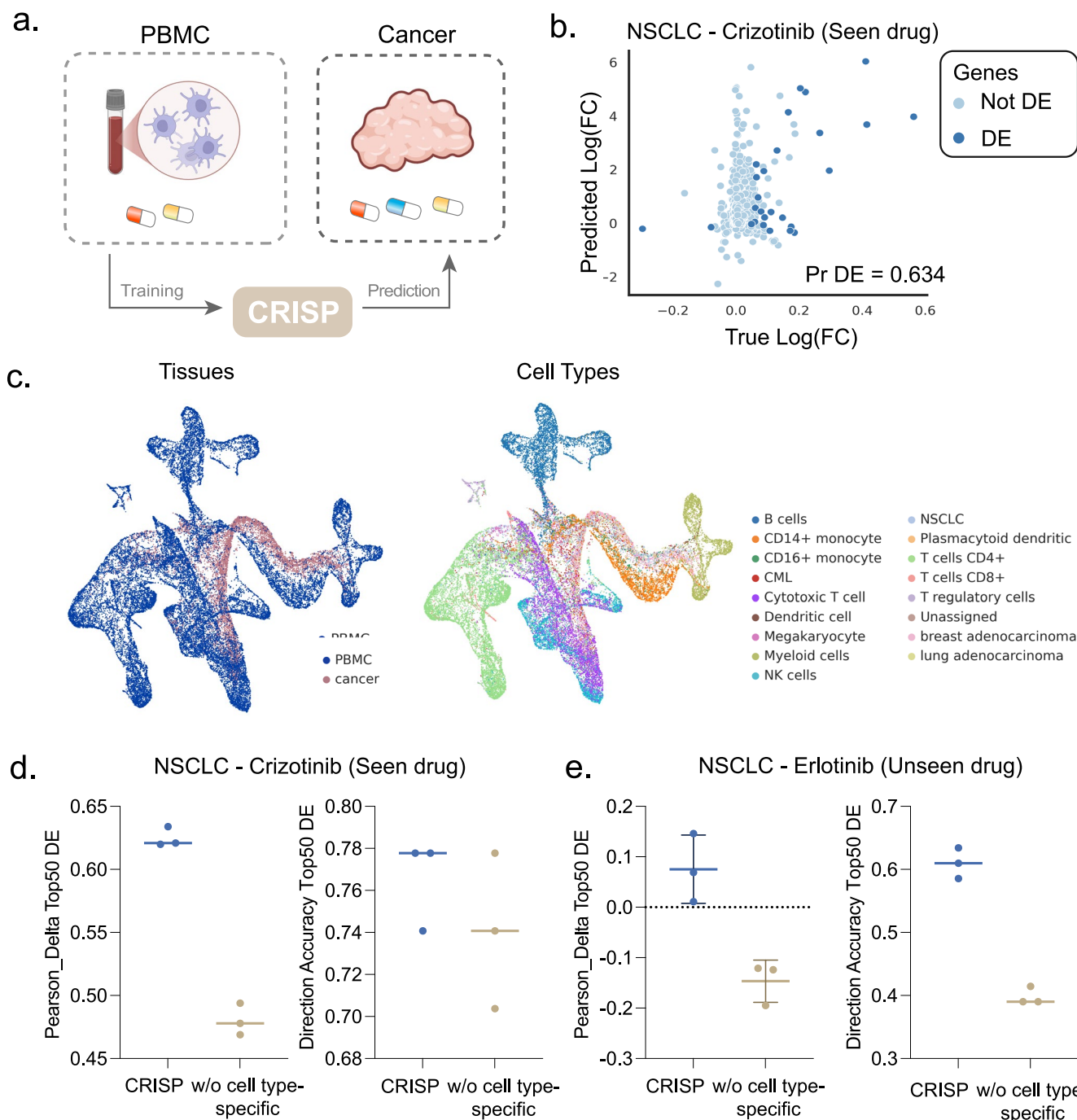
Peer review information *Nature Computational Science* thanks Jean Yang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Kaitlin McCardle, in collaboration with the *Nature Computational Science* team. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

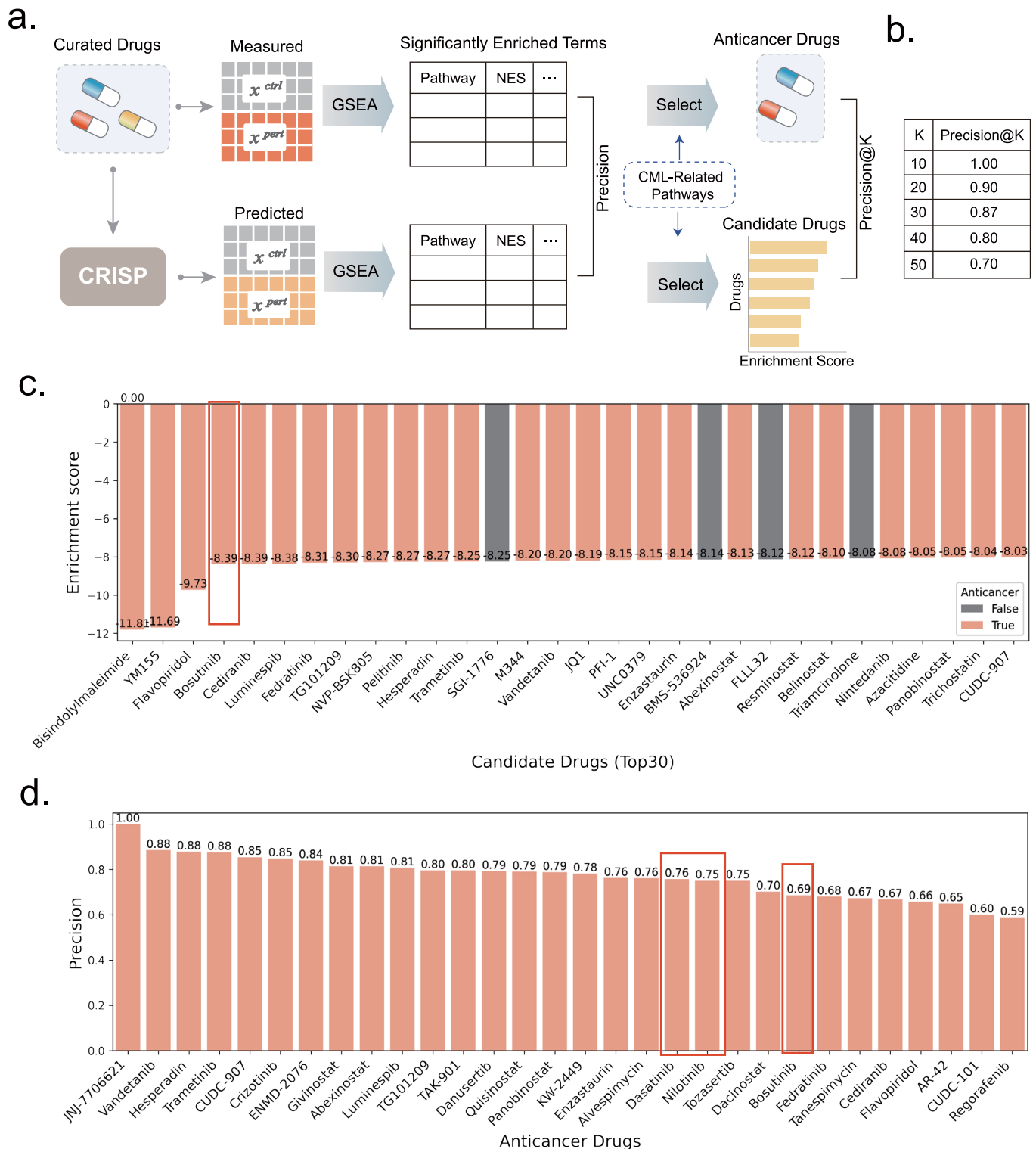
Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2025



Extended Data Fig. 1 | Prediction for cancer cells from the PBMC-trained CRISP. **a.** Illustration of cross-tissue prediction. We applied the trained CRISP from the NeurIPS dataset (PBMC) to cancer cells directly and predict the post-perturbation responses to both seen drug and unseen drug. **b.** Scatter plot showing the alignment of predicted log(FC) and observed log(FC), for Crizotinib-treated NSCLC. Crizotinib is a tyrosine kinase inhibitor that has approved for NSCLC treatment⁵⁷. **c.** UMAP visualization of predicted Crizotinib-treated samples, colored by tissues and cell types, respectively. It includes PBMC and cancer cell lines from the NeurIPS, PBMC-Bench, SciPlex3 and PC9 dataset.

d,e. Evaluation results for drug response prediction for NSCLC to the seen and unseen drug. We take CRISP without cell type-specific contrastive learning and classifier as baseline. Although it's not robust enough for unseen drug when predict across tissue, the addition of cell type-specific regularization makes progress. **f.** Comparison between predicted and observed log(FC) on top20 DE genes for Erlotinib-treated NSCLC. Each bar shows the averaged log(FC) on the corresponding gene. It reflects the predicted performance on direction and absolute value. Some icons are created with [BioRender.com](https://www.biorender.com).



Extended Data Fig. 2 | Drug screening on SciPlex3 dataset. **a.** We perform drug screening on all measured drugs in the SciPlex3 dataset, with the held-out K562 cell line (Chronic Myeloid Leukemia, CML) serving as the target for prediction. A set of CML-related pathways, which are significantly down-regulated upon treatment with approved CML drugs, is selected for screening and ground truth identification. Since nearly all significant pathways are down-regulated, we consider only negatively enriched terms. For real experimental data, a drug is classified as an anti-cancer drug (ground truth) if its significantly enriched down-regulated pathways overlap with the CML-related pathways. For CRISP's predictions, we rank all candidate drugs based on the sum of enrichment scores of CML-related pathways and compute Precision@K using the ground truth anti-

cancer drug labels. Additionally, we evaluate the precision of CRISP's predicted enriched terms by comparing them with real data for each drug. **b.** Precision@K for different values of K in the ranked list of candidate drugs. Notably, the top 10 ranked drugs are all labeled as anti-cancer drugs. **c.** The ranking of enrichment scores for the top 30 candidate drugs. Among them, only four are not anti-cancer drugs (labeled in gray). The approved CML drug 'Bosutinib' ranks 4th in our screening results. **d.** The precision of selected anti-cancer drugs, sorted by precision value. Specifically, the three approved CML drugs (highlighted with red frames)—'Dasatinib', 'Nilotinib', and 'Bosutinib'—achieve precision values of approximately 0.7 and rank within the top 30.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ A description of all covariates tested |
| ☒ | ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Cell Ranger V7.0.1 (--include-introns=false, --chemistry=auto, other setting as default).
Data analysis	numpy=1.26.4, an open-source python package for numerical calculations. torch=2.4.0, an open-source deep learning framework in python. pandas=2.2.3 an open-source python package for data analysis. matplotlib=3.9.3, an open-source python package for visualization. scikit-learn=1.5.2, an open-source python package for machine learning. scipy=1.13.1, an open-source python package for mathematics, science, and engineering. scanpy=1.10.3 an open-source python package for single-cell RNA sequencing (scRNA-seq) analysis. Single sample gene sets enrichment analysis 2.0 (ssGSEA2.0), open source R package (github: https://github.com/broadinstitute/ssGSEA2.0). CRISP our primary model code (https://github.com/ml4bio/CRISP) for benchmarking analysis of the comparison results.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

NeurIPS dataset (<https://www.kaggle.com/competitions/open-problems-single-cell-perturbations/data>). SciPlex3 dataset (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM4150378>). GBM dataset (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE148842>). PC9 dataset (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE134839>, <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE160244>). PBMC-Bench Dataset (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE132044>).

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

As this study utilized publicly available data rather than collecting primary data, demographic information regarding sex and gender was not reported. Furthermore, no specialized data filtering was applied during analysis, and the algorithm was designed to be gender-neutral, making it applicable to all individuals.

Population characteristics

Not applicable.

Recruitment

Not applicable.

Ethics oversight

Not applicable.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

We use all samples in the datasets. NeurIPS dataset measures the perturbation response across 18211 genes of 240059 cells from PBMC to 147 compositions, including 144 drugs with dosage as 1 μ M, Dimethylsulfoxide(DMSO) with dosage as 14.1 μ M as negative control and 2 drugs (dabrafenib and belinostat) with dosage as 0.1 μ M as positive control. It contains six cell types, including five lymphoid cell types: CD4+ T cells, CD8+ T cells, NK cells, T regulatory cells, B cells, and one myeloid cell type. For B cells and myeloid cells, only part of drug perturbation data is accessible. In the experiments, we treat all drugs except DMSO as the treatment group and samples with DMSO as the control group. Moreover, to be more comprehensive, the perturbation prediction is performed across all 18211 gene features without genes subset. SciPlex3 data contains the perturbation response of 581777 cells from 3 cancer cell lines: A549 human lung adenocarcinoma, MCF7 breast adenocarcinoma and K562 chronic myelogenous leukemia cells. There are 187 drug treatments, and single cell transcriptome measurement is performed in 58347 genes in total. GBM dataset contains measurements for 155377 cells from glioblastoma tissue with 6 drug treatments across 60725 genes. Samples of this dataset come from 9 different patients, we use GBM to perform the prediction across different unseen patients. PC9 dataset contains single-cell transcriptome measurements for non small cell lung cancer (NSCLC). In this study, we use two sub-datasets (GSE134839 and GSE160244) for evaluation, which contains Crizotinib- and Erlotinib-treated NSCLC samples. There are 4000 highly variable genes left after filtering. PBMC-Bench dataset contains scRNA-seq data of 31021 PBMC cells in unperturbed state from 7 sequencing methods: Drop-seq, inDrops, Seq-Well, 10x Chromium (v2), 10x Chromium (v2)A, 10x Chromium (v2) B, 10x Chromium (v3), CEL-Seq2 and Smart-seq2. Samples come from 9 cell types. The scRNA-seq is measured across 33694 genes.

Data exclusions

For the statistical analysis in Fig. 2f, cells with gene expression values beyond ± 2 standard deviations from the mean were excluded to mitigate the influence of outliers. All other analyses include the complete set of data points without any exclusion.

Replication

The performance of our model could be reproduced, and we also offer codes, and tutorials for the key results.

Randomization

We apply use different seed to initialize model's weight before training, and we can get consistent results for different seeds.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems		Methods	
n/a	Involved in the study	n/a	Involved in the study
☒	☐ Antibodies	☒	☐ ChIP-seq
☒	☐ Eukaryotic cell lines	☒	☐ Flow cytometry
☒	☐ Palaeontology and archaeology	☒	☐ MRI-based neuroimaging
☒	☐ Animals and other organisms		
☒	☐ Clinical data		
☒	☐ Dual use research of concern		